Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 5

Title: Sorting and Filtering for Sales Analysis using Power BI

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Sorting and Filtering for Sales Analysis using Power BI

GitHub Repository Link: *https://github.com/nidhidev2122/Business_analytics*

Problem Statement

A sales manager wants to identify the company's highest-value transactions and its underperforming products. Use sorting and filtering techniques in Power BI to extract the relevant business information from the sales data.

Dataset used: Sales Performance Analysis dataset (FY Oct 2025 – Sep 2026)

Source: sales_analysis_dataset.xlsx, a synthetic practice dataset of 300 transactions, 12 products and 4 regions (not real company data).

Steps Followed:

Step 1: Data Collection

The Sales Performance Analysis workbook was used for this experiment. The Sales_Data sheet holds one row per transaction with the following fields:

- Transaction ID, Date, Region, Salesperson, Customer
- Product and Category
- Quantity, Unit Price, Discount % and Unit Cost
- Revenue = Quantity × Unit Price × (1 – Discount %)
- Profit = Revenue – Quantity × Unit Cost, and a Returned flag (Y/N)

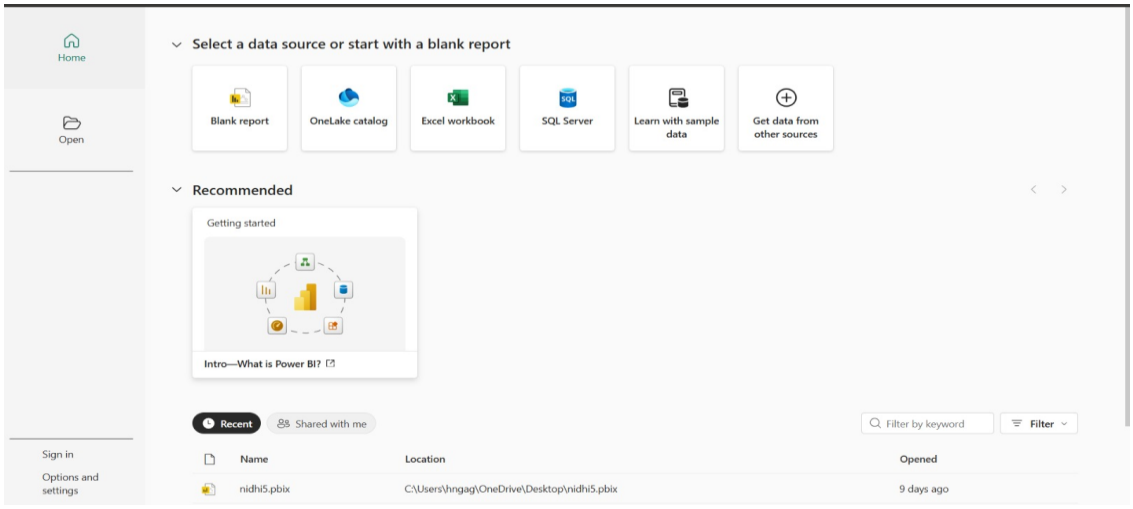


Figure 1: Power BI Desktop Home Screen

Step 2: Data Import

The workbook was loaded using Get Data > Excel Workbook and only the Sales_Data sheet was ticked, because the other sheets (Top_Transactions, High_Value, Product_Summary,

Underperformers) are pre-computed results that this experiment reproduces in Power BI. Transform Data was clicked to open Power Query Editor.

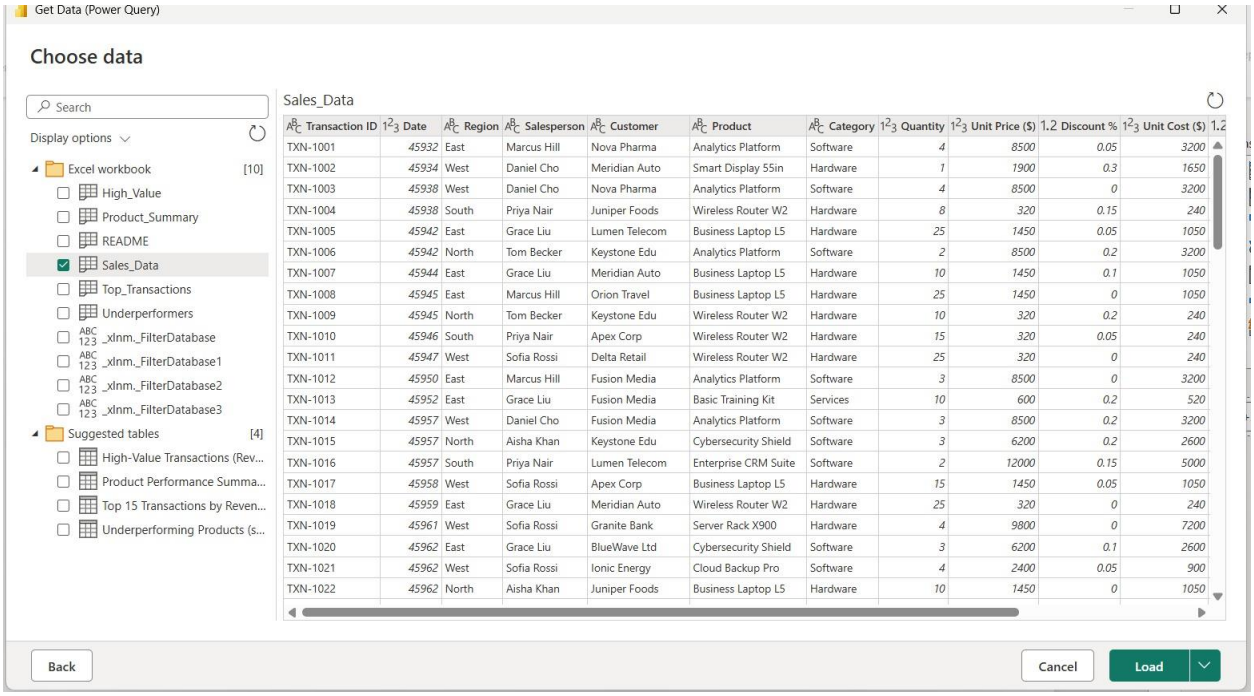


Figure 2: Get Data – showing Sales_Data selected

Step 3: Data Preparation

3.1 Data Types. Date was set to Date; Quantity to Whole Number; Unit Price, Discount %, Unit Cost, Revenue and Profit to Fixed Decimal Number; the remaining columns stayed as Text.

3.2 Data Quality Check. Column Quality, Column Distribution and Column Profile were enabled. All 14 columns showed 100% valid, 0% errors and 0% empty, and Transaction ID had 300 distinct and 300 unique values, so no cleaning or de-duplication was required.

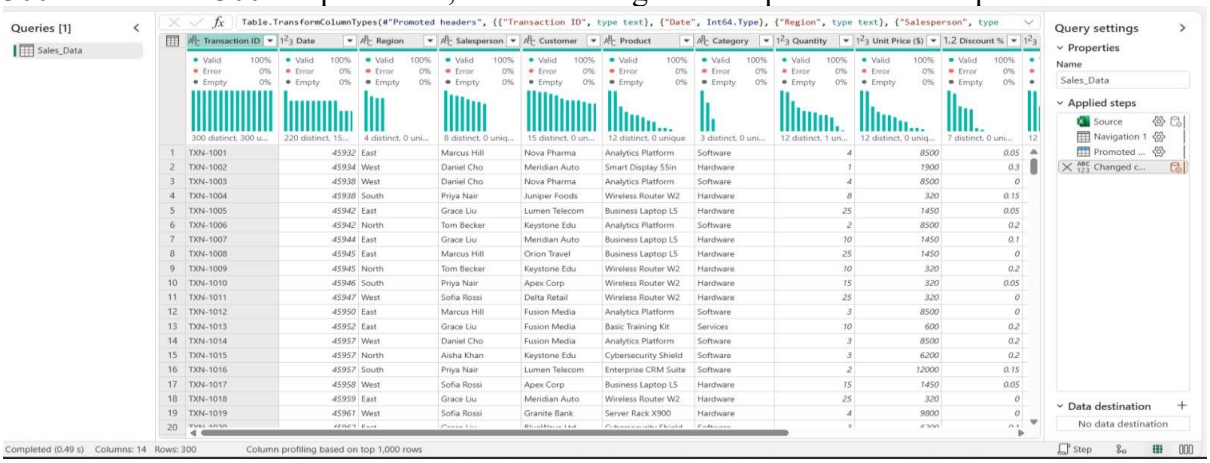


Figure 3: Power Query Editor – Column Quality and Applied Steps

Step 4: Sorting – Highest-Value Transactions

4.1 Sort by Revenue. In Power Query, the Revenue ($) column was sorted Descending, so the largest deals appear first.

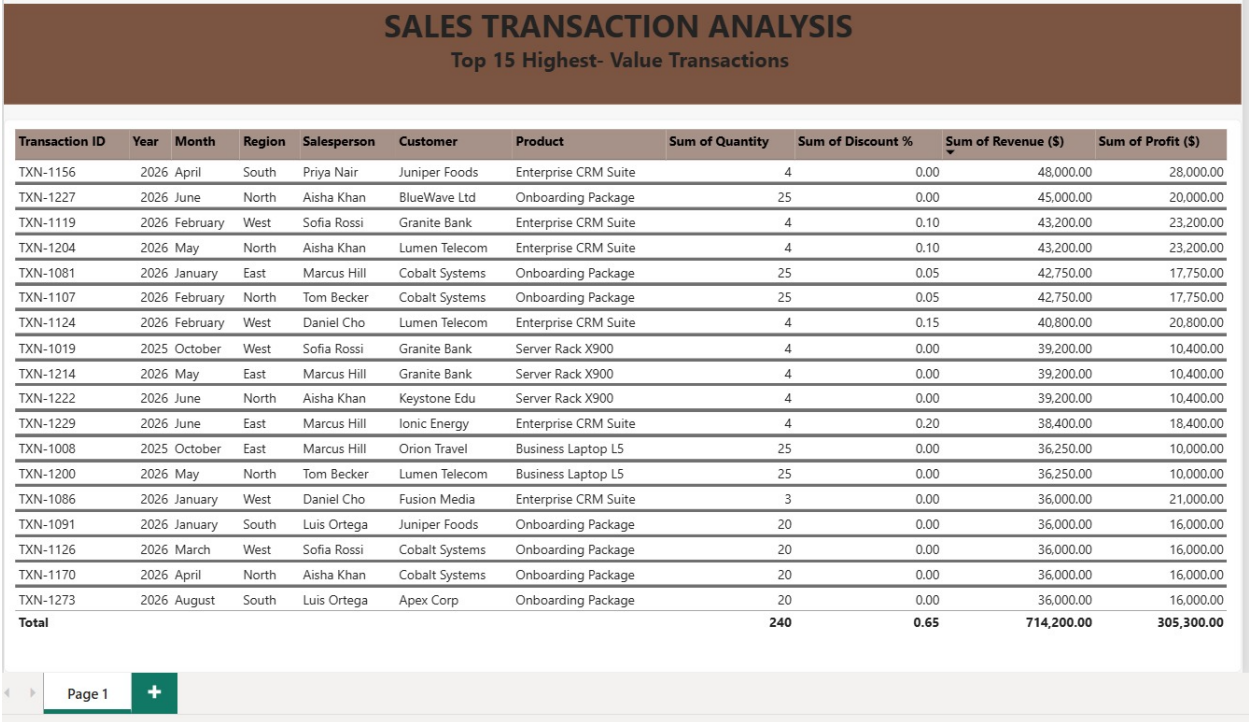
4.2 Top 15 Transactions. Keep Rows > Keep Top Rows > 15 was applied after the sort. In the report view the same result is achieved with a Table visual, using Filters pane > Transaction ID > Filter type: Top N > Top 15 > By value: Revenue ($).

```
Sorted  = Table.Sort(Typed, {{"Revenue ($)", Order.Descending}}),
Top15   = Table.FirstN(Sorted, 15)
```

Result: the 15 largest transactions together are worth $606,200, which is 14.1% of total revenue of $4,289,922 earned from only 5% of the transactions.

#	Txn ID	Region	Salesperson	Customer	Product	Qty	Disc.	Revenue	Profit
1	TXN-1156	South	Priya Nair	Juniper Foods	Enterprise CRM Suite	4	0%	$48,000	$28,000
2	TXN-1227	North	Aisha Khan	BlueWave Ltd	Onboarding Package	25	0%	$45,000	$20,000
3	TXN-1204	North	Aisha Khan	Lumen Telecom	Enterprise CRM Suite	4	10%	$43,200	$23,200
4	TXN-1119	West	Sofia Rossi	Granite Bank	Enterprise CRM Suite	4	10%	$43,200	$23,200
5	TXN-1107	North	Tom Becker	Cobalt Systems	Onboarding Package	25	5%	$42,750	$17,750
6	TXN-1081	East	Marcus Hill	Cobalt Systems	Onboarding Package	25	5%	$42,750	$17,750
7	TXN-1124	West	Daniel Cho	Lumen Telecom	Enterprise CRM Suite	4	15%	$40,800	$20,800
8	TXN-1222	North	Aisha Khan	Keystone Edu	Server Rack X900	4	0%	$39,200	$10,400
9	TXN-1214	East	Marcus Hill	Granite Bank	Server Rack X900	4	0%	$39,200	$10,400
10	TXN-1019	West	Sofia Rossi	Granite Bank	Server Rack X900	4	0%	$39,200	$10,400
11	TXN-1229	East	Marcus Hill	Ionic Energy	Enterprise CRM Suite	4	20%	$38,400	$18,400
12	TXN-1200	North	Tom Becker	Lumen Telecom	Business Laptop L5	25	0%	$36,250	$10,000
13	TXN-1008	East	Marcus Hill	Orion Travel	Business Laptop L5	25	0%	$36,250	$10,000
14	TXN-1273	South	Luis Ortega	Apex Corp	Onboarding Package	20	0%	$36,000	$16,000
15	TXN-1170	North	Aisha Khan	Cobalt Systems	Onboarding Package	20	0%	$36,000	$16,000

Table 1: Top 15 transactions sorted by Revenue (descending)



Transaction ID	Year	Month	Region	Salesperson	Customer	Product	Sum of Quantity	Sum of Discount %	Sum of Revenue ($)	Sum of Profit ($)
TXN-1156	2026	April	South	Priya Nair	Juniper Foods	Enterprise CRM Suite	4	0.00	48,000.00	28,000.00
TXN-1227	2026	June	North	Aisha Khan	BlueWave Ltd	Onboarding Package	25	0.00	45,000.00	20,000.00
TXN-1119	2026	February	West	Sofia Rossi	Granite Bank	Enterprise CRM Suite	4	0.10	43,200.00	23,200.00
TXN-1204	2026	May	North	Aisha Khan	Lumen Telecom	Enterprise CRM Suite	4	0.10	43,200.00	23,200.00
TXN-1081	2026	January	East	Marcus Hill	Cobalt Systems	Onboarding Package	25	0.05	42,750.00	17,750.00
TXN-1107	2026	February	North	Tom Becker	Cobalt Systems	Onboarding Package	25	0.05	42,750.00	17,750.00
TXN-1124	2026	February	West	Daniel Cho	Lumen Telecom	Enterprise CRM Suite	4	0.15	40,800.00	20,800.00
TXN-1019	2025	October	West	Sofia Rossi	Granite Bank	Server Rack X900	4	0.00	39,200.00	10,400.00
TXN-1214	2026	May	East	Marcus Hill	Granite Bank	Server Rack X900	4	0.00	39,200.00	10,400.00
TXN-1222	2026	June	North	Aisha Khan	Keystone Edu	Server Rack X900	4	0.00	39,200.00	10,400.00
TXN-1229	2026	June	East	Marcus Hill	Ionic Energy	Enterprise CRM Suite	4	0.20	38,400.00	18,400.00
TXN-1008	2025	October	East	Marcus Hill	Orion Travel	Business Laptop L5	25	0.00	36,250.00	10,000.00
TXN-1200	2026	May	North	Tom Becker	Lumen Telecom	Business Laptop L5	25	0.00	36,250.00	10,000.00
TXN-1086	2026	January	West	Daniel Cho	Fusion Media	Enterprise CRM Suite	3	0.00	36,000.00	21,000.00
TXN-1091	2026	January	South	Luis Ortega	Juniper Foods	Onboarding Package	20	0.00	36,000.00	16,000.00
TXN-1126	2026	March	West	Sofia Rossi	Cobalt Systems	Onboarding Package	20	0.00	36,000.00	16,000.00
TXN-1170	2026	April	North	Aisha Khan	Cobalt Systems	Onboarding Package	20	0.00	36,000.00	16,000.00
TXN-1273	2026	August	South	Luis Ortega	Apex Corp	Onboarding Package	20	0.00	36,000.00	16,000.00
Total							**240**	**0.65**	**714,200.00**	**305,300.00**

Figure 4: Top 15 transactions – Table visual with Top N filter

Step 5: Filtering – High-Value Transactions

5.1 Defining "high-value". A high-value transaction was defined as one in the top 10% by revenue. The 90th percentile of Revenue is $32,400, calculated with the DAX measure below.

```
High Value Threshold = PERCENTILE.INC(Sales_Data[Revenue ($)], 0.9)
```

5.2 Applying the filter. Rows were filtered with Revenue ($) greater than or equal to 32,400 (Power Query: Number Filters > Greater Than Or Equal To). In a visual, the Revenue field must be set to "Don't summarize" first, otherwise the filter acts on totals instead of individual transactions.

```
HighValue = Table.SelectRows(Sorted, each [#"Revenue ($)"] >= 32400)
```

Result: 32 transactions qualified, worth $1,187,173 (27.7% of all revenue) and $477,573 of profit. Only 1 of the 32 was returned.

Product	High-value txns	Revenue	Share of high-value
Enterprise CRM Suite	10	$379,200	31.9%
Business Laptop L5	8	$277,313	23.4%
Onboarding Package	7	$274,500	23.1%
Server Rack X900	5	$188,160	15.8%
Analytics Platform	2	$68,000	5.7%

Table 2: High-value transactions by product

Region	High-value txns	Revenue	Share of high-value
West	12	$430,118	36.2%

Region	High-value txns	Revenue	Share of high-value
North	8	$307,425	25.9%
East	6	$225,475	19.0%
South	6	$224,155	18.9%

Table 3: High-value transactions by region

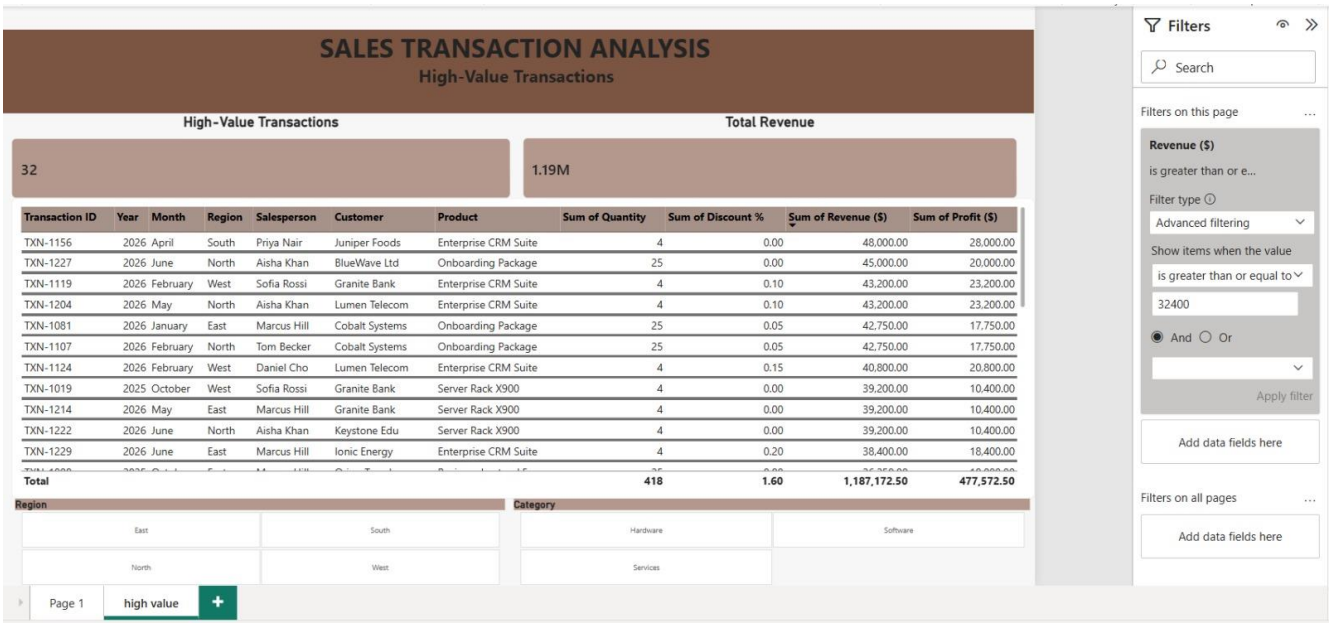


Figure 5: High-value transactions (Revenue ≥ $32,400) with Region and Category slicers

Step 6: Filtering – Underperforming Products

6.1 Product summary. Transactions were grouped by Product to get transactions, units, revenue, profit and margin per product (Home > Group By in Power Query, or the DAX measures below).

```
Total Revenue = SUM(Sales_Data[Revenue ($)])
Total Profit  = SUM(Sales_Data[Profit ($)])
Profit Margin % = DIVIDE([Total Profit], [Total Revenue])
Avg Product Revenue = AVERAGEX(ALL(Sales_Data[Product]), [Total Revenue])
Status = IF([Total Revenue] < 0.6 * [Avg Product Revenue]
           || [Profit Margin %] < 0.15, "Underperforming", "OK")
```

6.2 Defining "underperforming". A product was flagged if its revenue was below 60% of the average product revenue ($357,494, so the cut-off is $214,496) or its profit margin was below 15%. These thresholds are assumptions and can be changed.

6.3 Filtering and sorting. Rows with Status = Underperforming were kept and sorted by Revenue in ascending order, so the weakest product is first.

```
Under  = Table.SelectRows(Flagged, each [Status] = "Underperforming"),
Sorted = Table.Sort(Under, {{"Revenue", Order.Ascending}})
```

Product	Category	Txns	Units	Revenue	Profit	Margin	Rev. vs avg
Legacy Fax Gateway	Hardware	3	22	$7,943	-$1,078	-13.6%	2.2%

Product	Category	Txns	Units	Revenue	Profit	Margin	Rev. vs avg
Basic Training Kit	Services	5	61	$26,970	-$4,750	-17.6%	7.5%
Smart Display 55in	Hardware	5	24	$33,155	-$6,445	-19.4%	9.3%

Table 4: Underperforming products (lowest revenue first)

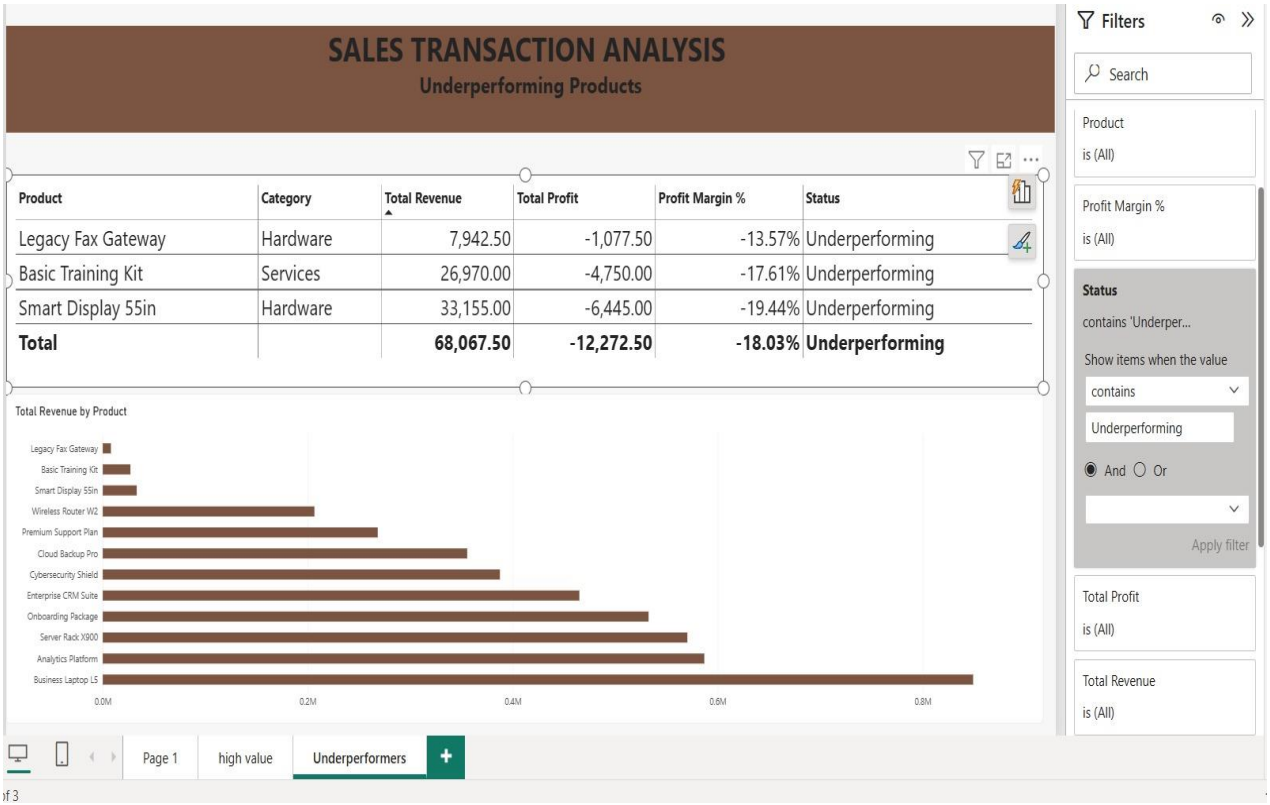
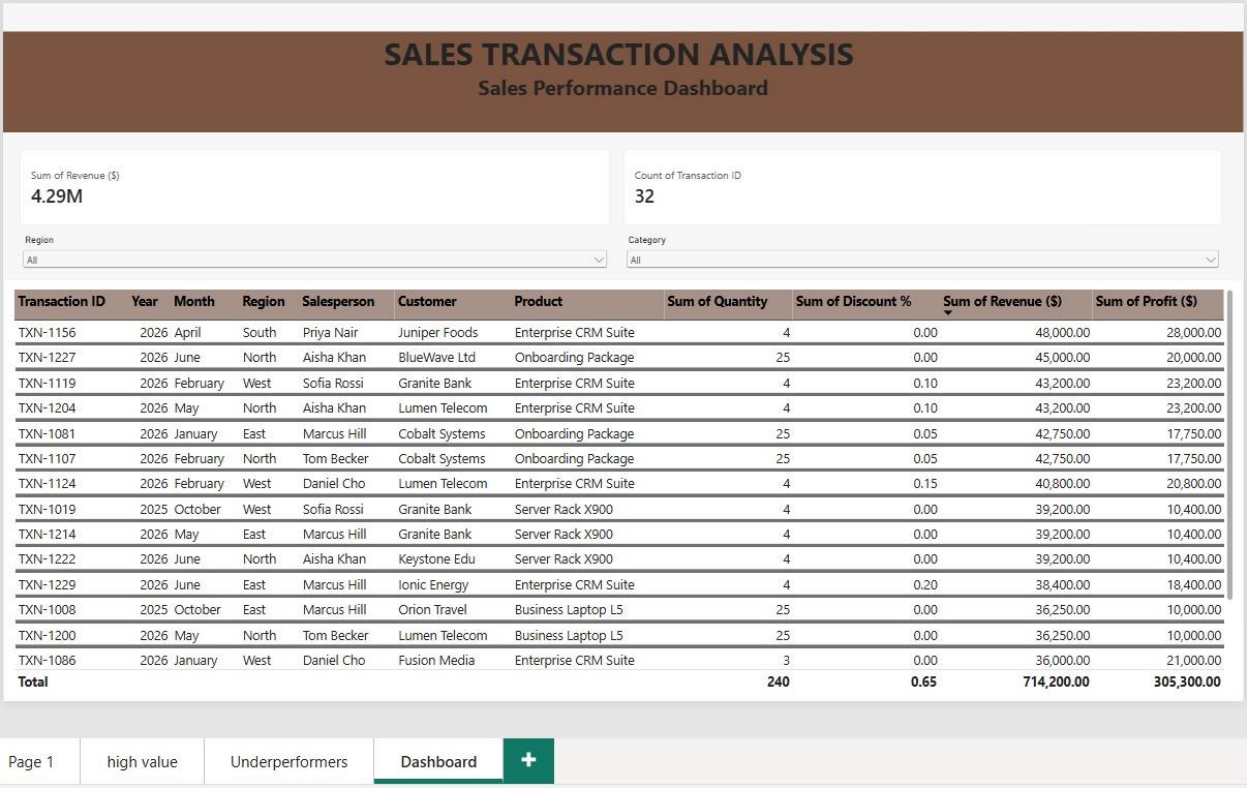


Figure 6: Underperforming products – bar chart and table



Transaction ID	Year	Month	Region	Salesperson	Customer	Product	Sum of Quantity	Sum of Discount %	Sum of Revenue ($)	Sum of Profit ($)
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TXN-1086	2026	January	West	Daniel Cho	Fusion Media	Enterprise CRM Suite	3	0.00	36,000.00	21,000.00
Total							**240**	**0.65**	**714,200.00**	**305,300.00**

Figure 7: Final Power BI dashboard

Summary:

- **Data quality:** All 300 transactions were complete and unique, with no missing values or errors.
- **Sorting:** Sorting Revenue in descending order showed the top deal at $48,000 (TXN-1156, Enterprise CRM Suite, South).
- **Filtering (high value):** 32 transactions at or above $32,400 make up 27.7% of revenue.
- **Filtering (underperformers):** 4 of 12 products were flagged: Legacy Fax Gateway, Basic Training Kit, Smart Display 55in and Wireless Router W2.

Conclusion:

This experiment applied sorting, Top N filtering, threshold filtering and grouped summaries in Power BI to answer two questions a sales manager asks: where the biggest deals come from, and which products are not pulling their weight. Sorting by revenue isolated the top 15 deals, a percentile-based filter defined 32 high-value transactions, and a revenue and margin rule flagged 4 underperforming products. Together these steps turned 300 raw transactions into a short, actionable list.

Business Insights:

- **Few deals carry a lot of revenue:** 10.7% of transactions (32 of 300) produce 27.7% of revenue.

- **Enterprise CRM Suite leads big deals:** 10 of the 32 high-value transactions ($379,200). Business Laptop L5 (8), Onboarding Package (7) and Server Rack X900 (5) follow. Only 14 CRM transactions exist in total, so most of them are large deals.
- **West region and Sofia Rossi stand out:** West has 12 of the 32 high-value deals ($430,118), and Sofia Rossi alone closed 9 ($319,318).
- **Discounts are linked to losses:** Legacy Fax Gateway, Basic Training Kit and Smart Display 55in all lose money, with average discounts of 22%, 27% and 26% against about 8% for most other products. Their list prices barely exceed unit cost, so discounts push them below cost.
- **Wireless Router W2 is a different case:** it is flagged on revenue only (57.8% of average), but it is profitable (18.0% margin) and the second most-sold product by units (706). It is a low-ticket, high-volume item (price $320) and probably does not need to be cut.
- **Top-selling does not mean most profitable:** Business Laptop L5 has the highest revenue ($849,193) but only a 21.1% margin, while Analytics Platform earns a 58.6% margin on $586,925.